6061 Introduction to Deep Learning

Exercise 3

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This exercise sheet will not be graded and is provided for practice only.

1. **Gradient descent.**
   (a) Starting from $(x_0, y_0) = (0, 1)$ perform two steps of gradient descent with step size/learning rate $\alpha = 0.1$ for

   $$f : \mathbb{R}^2 \to \mathbb{R}, f(x, y) = 3xy - 5y^2$$

   (b) Starting from $(x_0, y_0) = (1, 1)$ do one step of gradient descent with step size/learning rate $\alpha = 1$ for

   $$g(x, y) = xy^2 - x^3 \exp(x - y) + \sin^2(xy), g : \mathbb{R}^2 \to \mathbb{R}.$$

   (c) How can you modify the gradient descent algorithm into a gradient ascent algorithm that maximizes a function instead of minimizing it?

2. **Regressor evaluation.** Calculate mean absolute error (MAE) and mean squared error (MSE) for both models on the following data. Which model is better?

y	$\hat{y}^{(1)}$ (Model 1)	$\hat{y}^{(2)}$ (Model 2)
5	5	4.5
7	7	7.5
9	9	10
3	2	2.5
10	12	11

3. **Binary classifier evaluation.**
   (a) Calculate precision, recall, F1-score, and accuracy for the following data.

		Predicted	
		$\hat{y} = 0$	$\hat{y} = 1$
Ground truth	$y = 0$	3	147
	$y = 1$	17	137

   (b) Calculate precision, recall, and accuracy for the following data.

   $$y = [0, 1, 1, 1, 0, 1, 0, 1, 0, 1, 0]$$
   $$\hat{y} = [1, 0, 1, 1, 1, 1, 1, 0, 0, 0, 0]$$

4. **Multi-class classifier evaluation.** Now, a classifier predicts one of C classes. Such a classifier can be evaluated with the binary metrics by treating each class separately (one-vs-rest): to compute the metrics for class $y = 0$, we merge all other classes into a single negative class and add up their counts. In some other example, $\text{Recall}_{y=0} = \frac{3}{3+(147+33)} \approx 0.016$, where $(147 + 33)$ are the combined counts of the other classes.
   (a) Calculate precision, recall, and F1-score for each class. Then compute the so-called *macro F1-score* by averaging over all classes. Also compute accuracy.

		Predicted		
		$\hat{y} = 0$	$\hat{y} = 1$	$\hat{y} = 2$
Ground truth	$y = 0$	7	129	46
	$y = 1$	25	766	170
	$y = 2$	73	73	204

   (b) Discuss whether the classifier has good performance.